CHAPTER 1

An Introduction to the Human Body

Anatomy, physiology, levels of organization, homeostasis, and the language of the body.

Anatomy & Physiology · Broward-Miami Health Institute

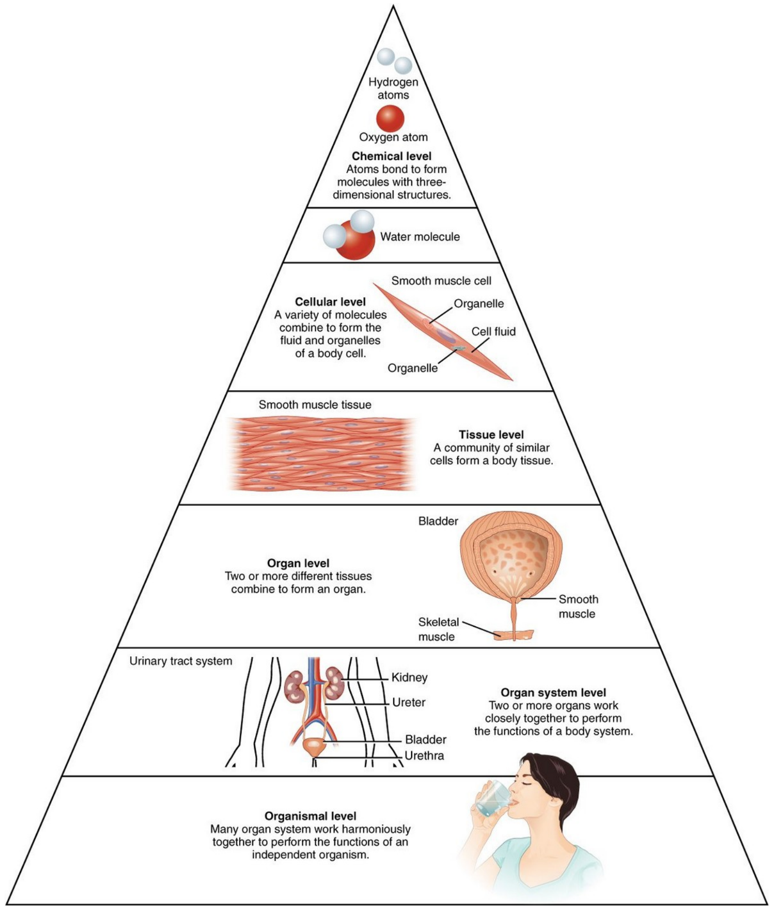
FACULTY EDITION

CHAPTER 1

Learning Objectives

After studying this chapter, students will be able to:

1. Distinguish between anatomy and physiology, and identify several branches of each
2. Describe the structure of the body, from simplest to most complex, in terms of the six levels of organization
3. Identify the functional characteristics of human life
4. Identify the four requirements for human survival
5. Define homeostasis and explain its importance to normal human functioning
6. Use appropriate anatomical terminology to identify key body structures, body regions, and directions in the body



The body's levels of organization — the roadmap for this chapter.

1.1

SECTION

Overview of Anatomy and Physiology

The study of the body's structure and how its parts function — and why the two are inseparable.

SECTION 1.1

Anatomy & Physiology in Action

- Anatomy = structure; physiology = function
- Form and function are always linked
- Gross (macroscopic) vs. microscopic anatomy
- A blood-pressure reading is physiology at the bedside

Measuring blood pressure — physiology you assess every shift

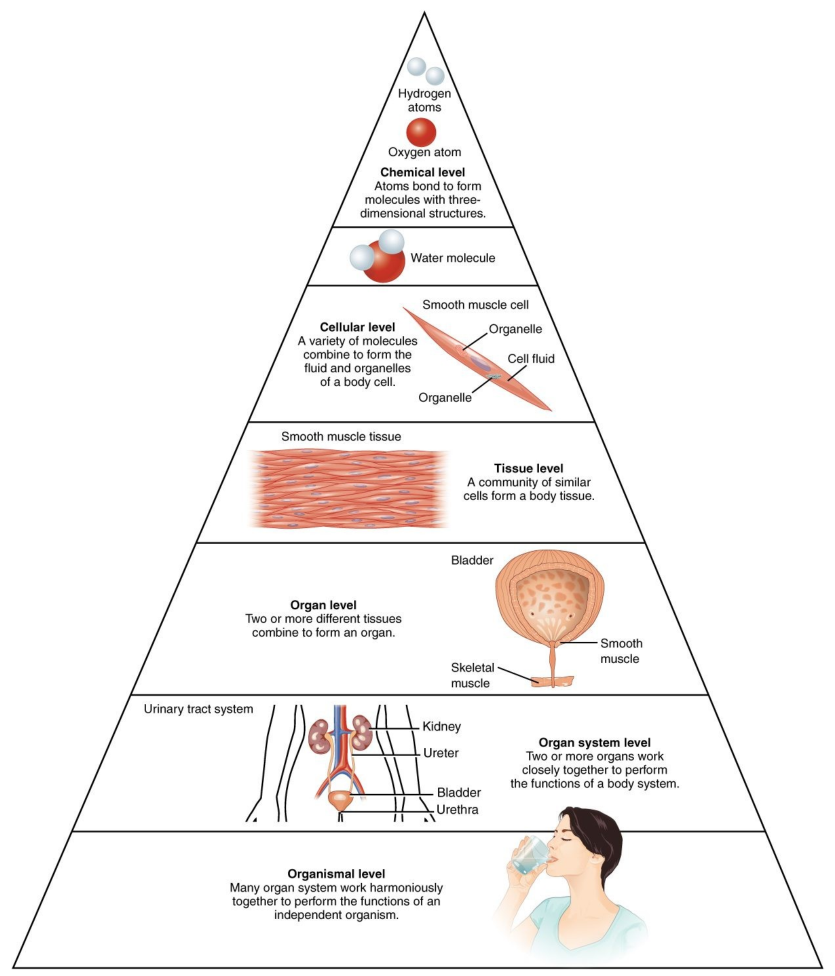
1.2

SECTION

Structural Organization of the Human Body

From atoms to the whole organism — six levels of increasing complexity, and the eleven organ systems.

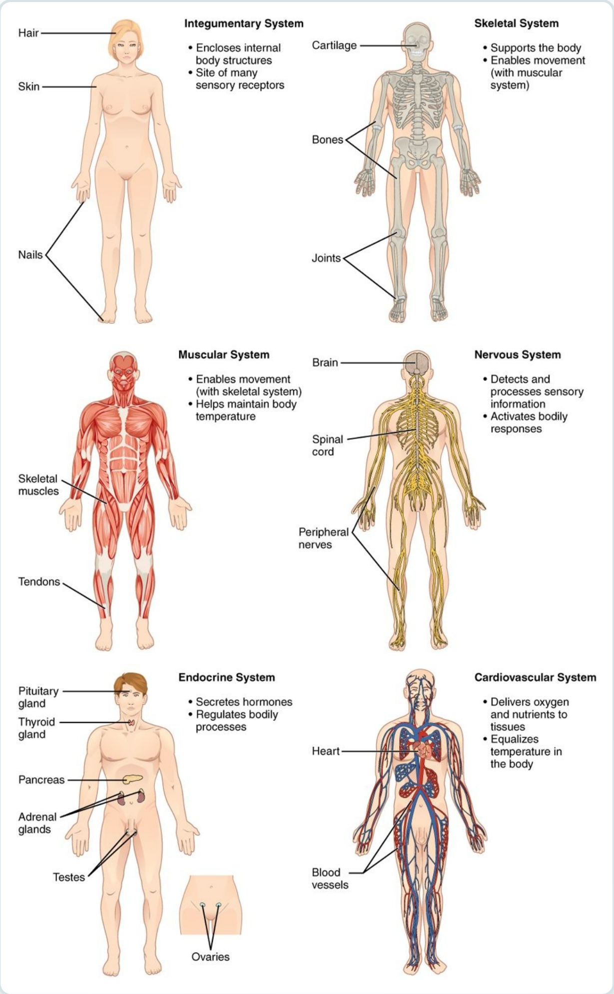
The Six Levels of Organization



The levels of structural organization in the body

- Chemical → cellular → tissue → organ → system → organism
- Each level builds on the one below it
- The cell is the smallest unit of life
- Complexity rises from atom to whole person

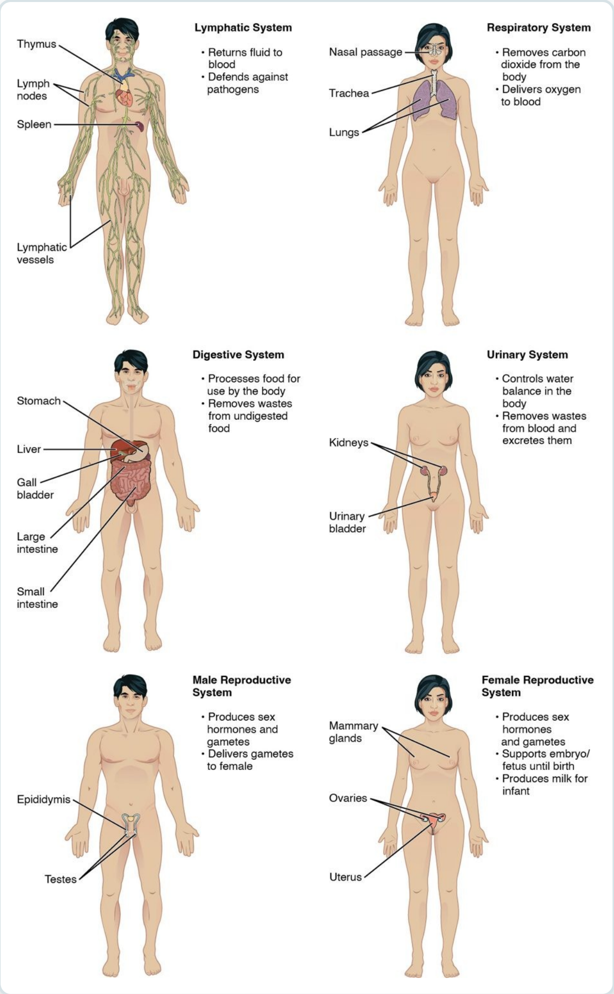
Organ Systems of the Body (I)



Organ systems and their major organs (part 1)

- Integumentary, skeletal, and muscular systems
- Nervous and endocrine — the body's control systems
- Cardiovascular system transports blood
- Each system has distinct organs and roles

Organ Systems of the Body (II)



Organ systems and their major organs (part 2)

- Lymphatic/immune and respiratory systems
- Digestive and urinary systems
- Male and female reproductive systems
- All eleven systems work interdependently

1.3

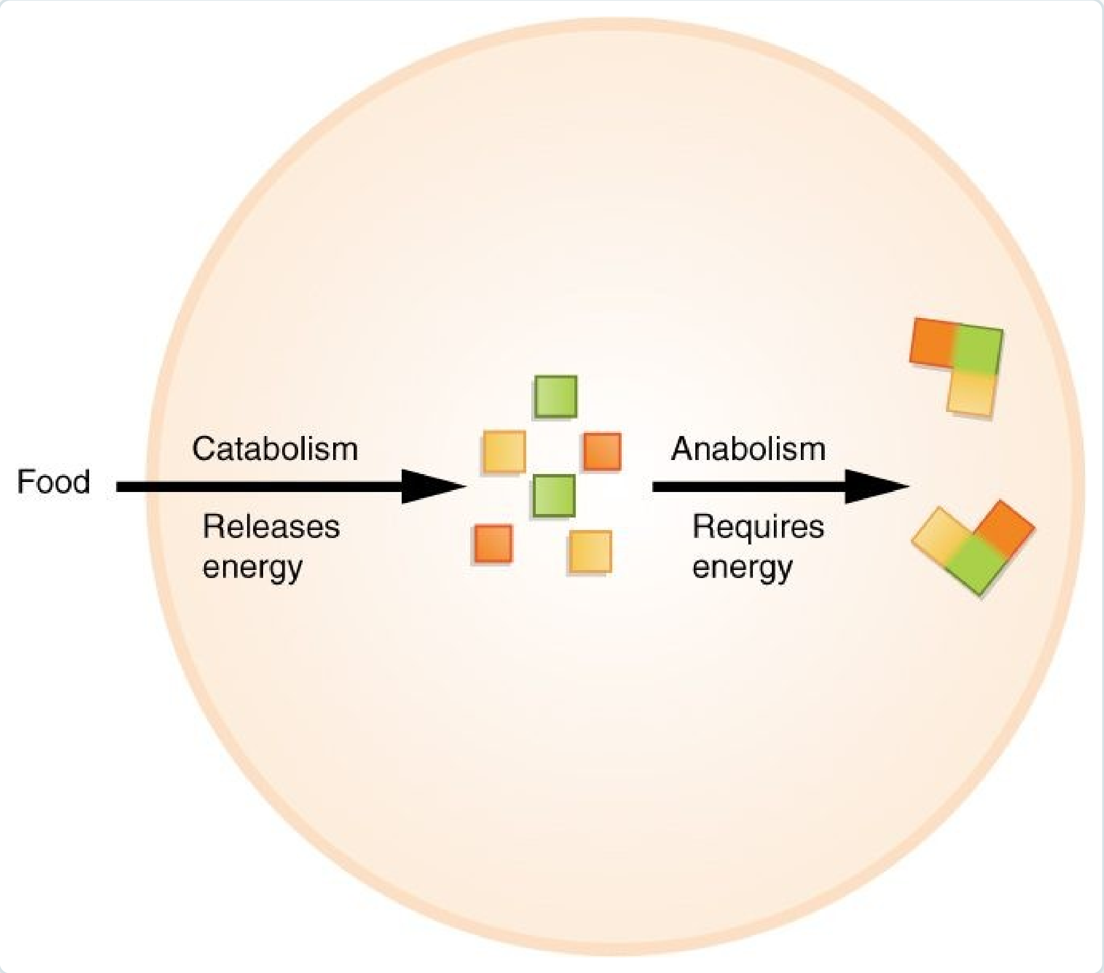
SECTION

Functions of Human Life

The processes that distinguish the living human body — organization, metabolism, responsiveness, and more.

Functions of Human Life

- Organization, metabolism, responsiveness
- Movement, development, and reproduction
- Metabolism = catabolism + anabolism
- Catabolism releases energy; anabolism builds



Metabolism: catabolism releases energy, anabolism uses it

1.4

SECTION

Requirements for Human Life

Oxygen, nutrients, water, and narrow ranges of temperature and atmospheric pressure.

SECTION 1.4

Requirements for Human Survival

- Oxygen, nutrients, and water
- Narrow ranges of temperature and pressure
- Demand rises sharply during activity
- Survival depends on meeting all of these

The body's demands climb steeply with activity

1.5

SECTION

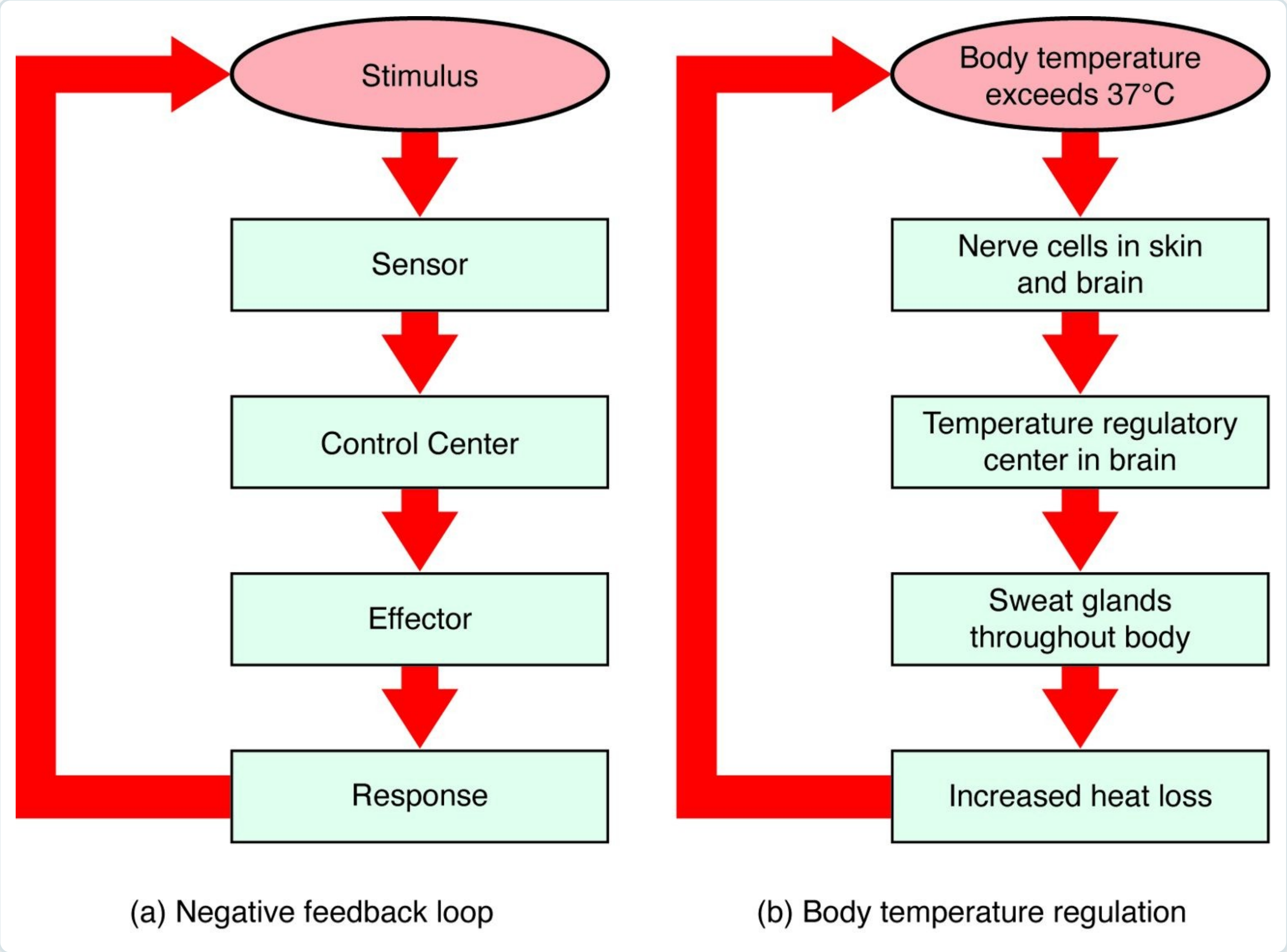
Homeostasis

How the body maintains a stable internal environment through feedback.

SECTION 1.5

Homeostasis & Negative Feedback

- Homeostasis = a stable internal environment
- Negative feedback reverses a change
- Stimulus → sensor → control center → effector
- Example: temperature regulation by sweating

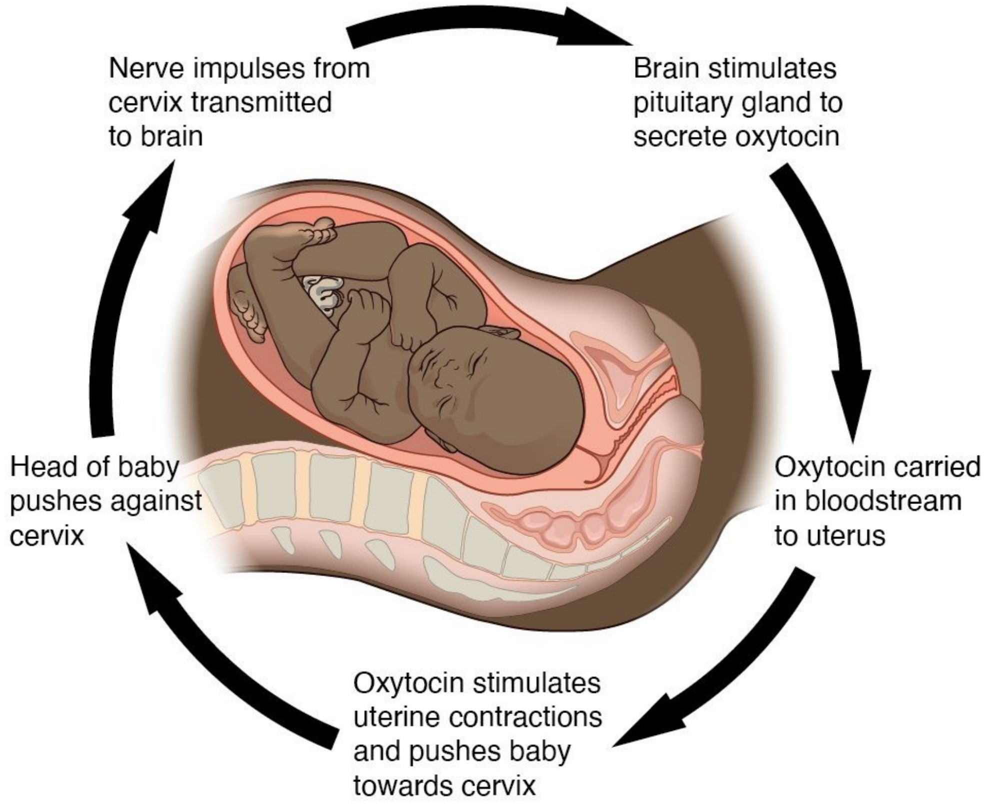


The negative-feedback loop and body-temperature control

SECTION 1.5

Positive Feedback

- Amplifies a change rather than reversing it
- Reserved for rapid, self-limiting events
- Example: oxytocin during labor and childbirth
- Also drives the blood-clotting cascade



Positive feedback: oxytocin intensifies labor contractions

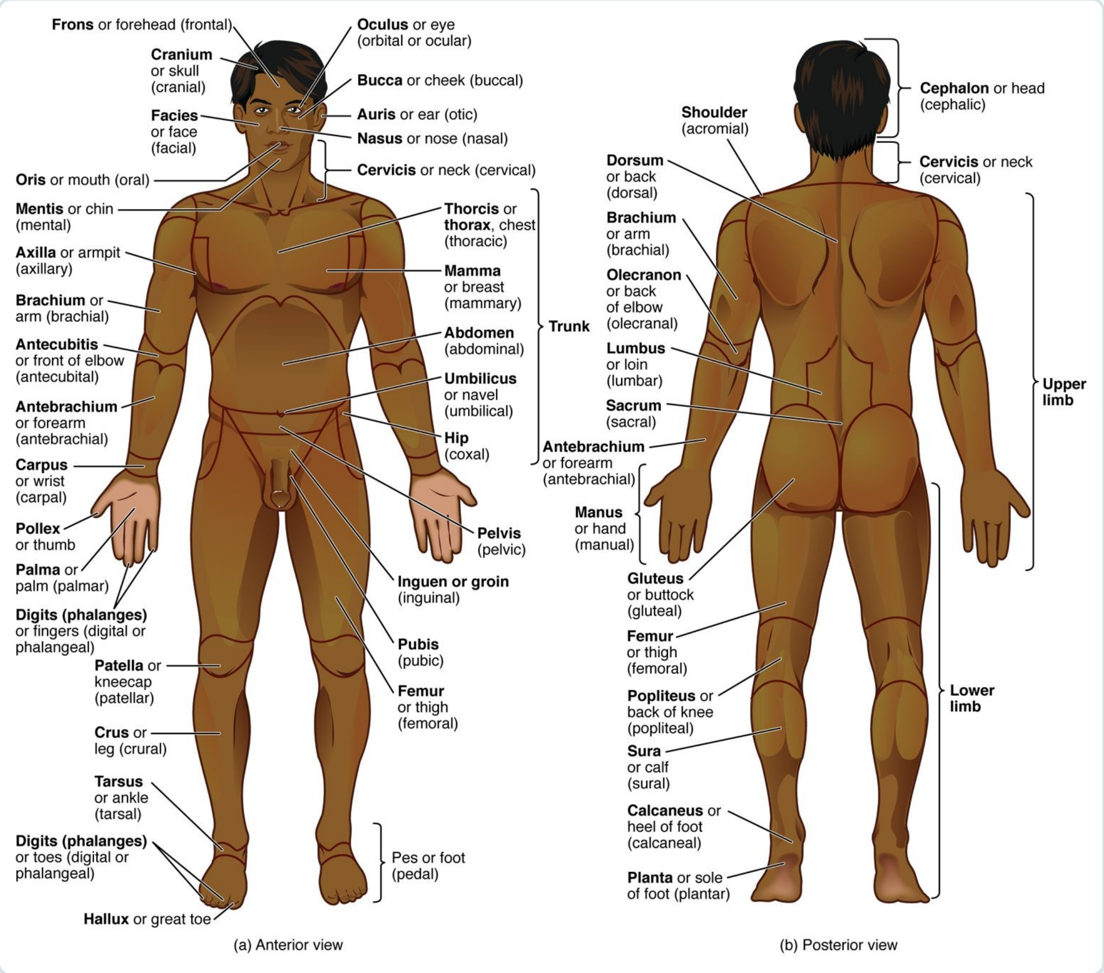
SECTION

1.6 Anatomical Terminology

The standardized language of position, direction, planes, cavities, and regions.

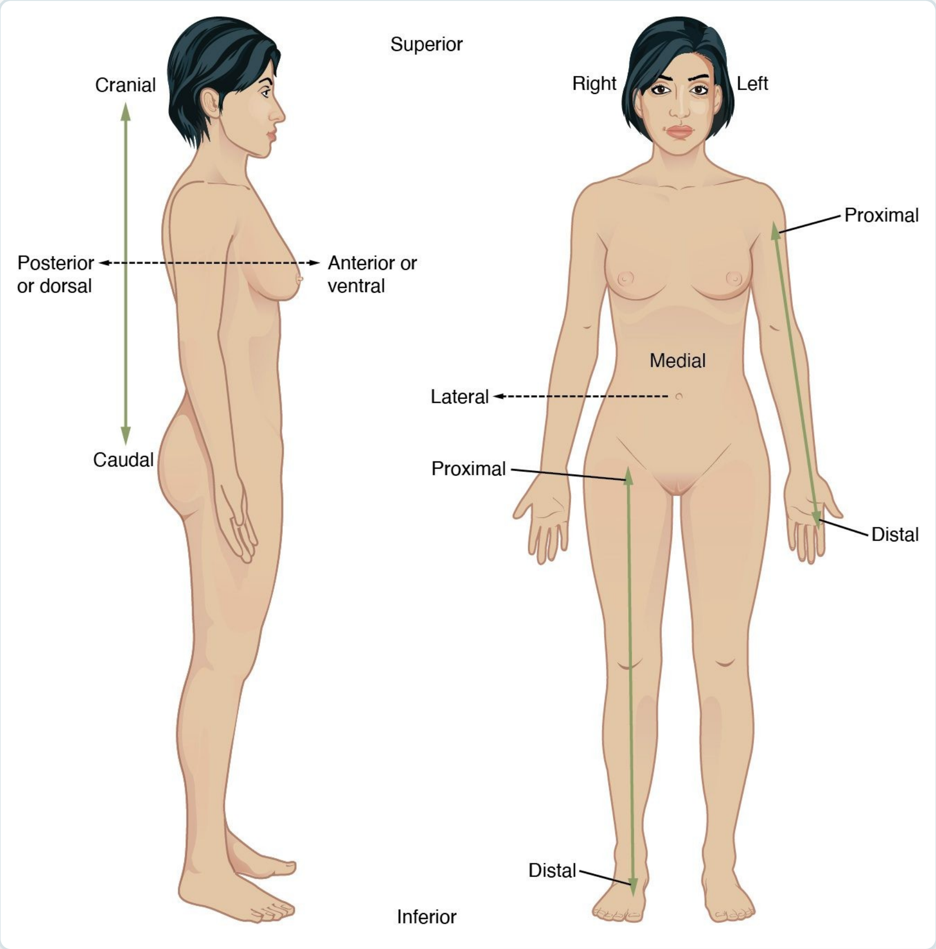
Anatomical Position & Body Regions

- Anatomical position is the reference posture
- Standing, facing forward, palms forward
- Regional terms name specific body areas
- A shared language prevents miscommunication



Anatomical position with regional terms (anterior and posterior)

Directional Terms

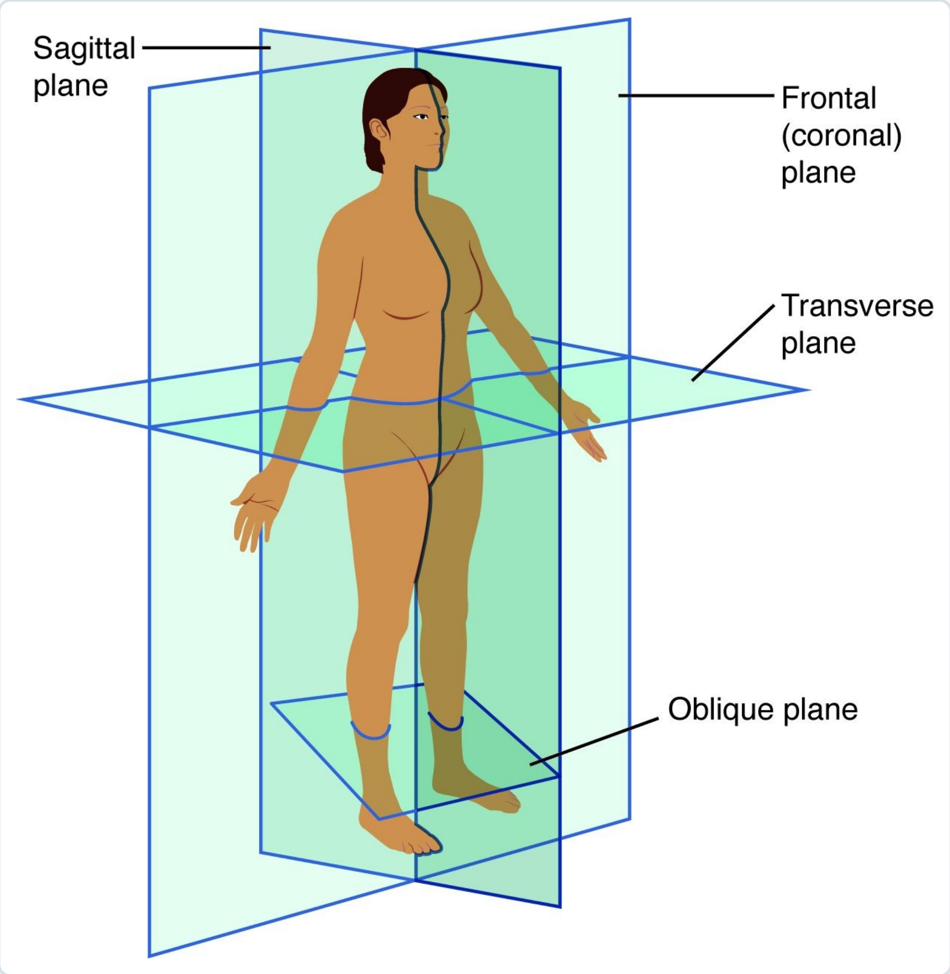


Directional terms used to relate body structures

- Superior/inferior, anterior/posterior
- Medial/lateral, proximal/distal
- Superficial/deep
- Always described from anatomical position

Planes of Section

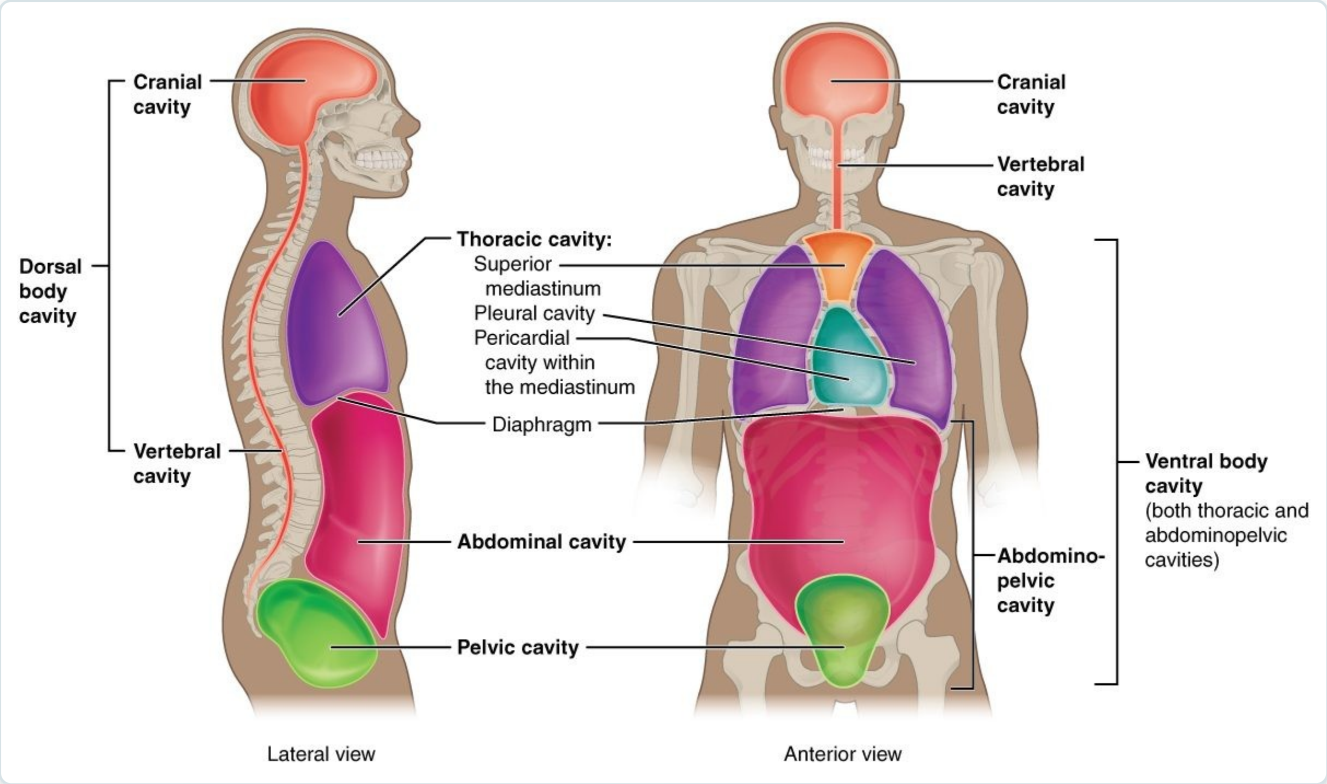
- Sagittal plane divides left and right
- Frontal (coronal) divides front and back
- Transverse divides superior and inferior
- Imaging studies are read in these planes



The sagittal, frontal, and transverse planes

SECTION 1.6

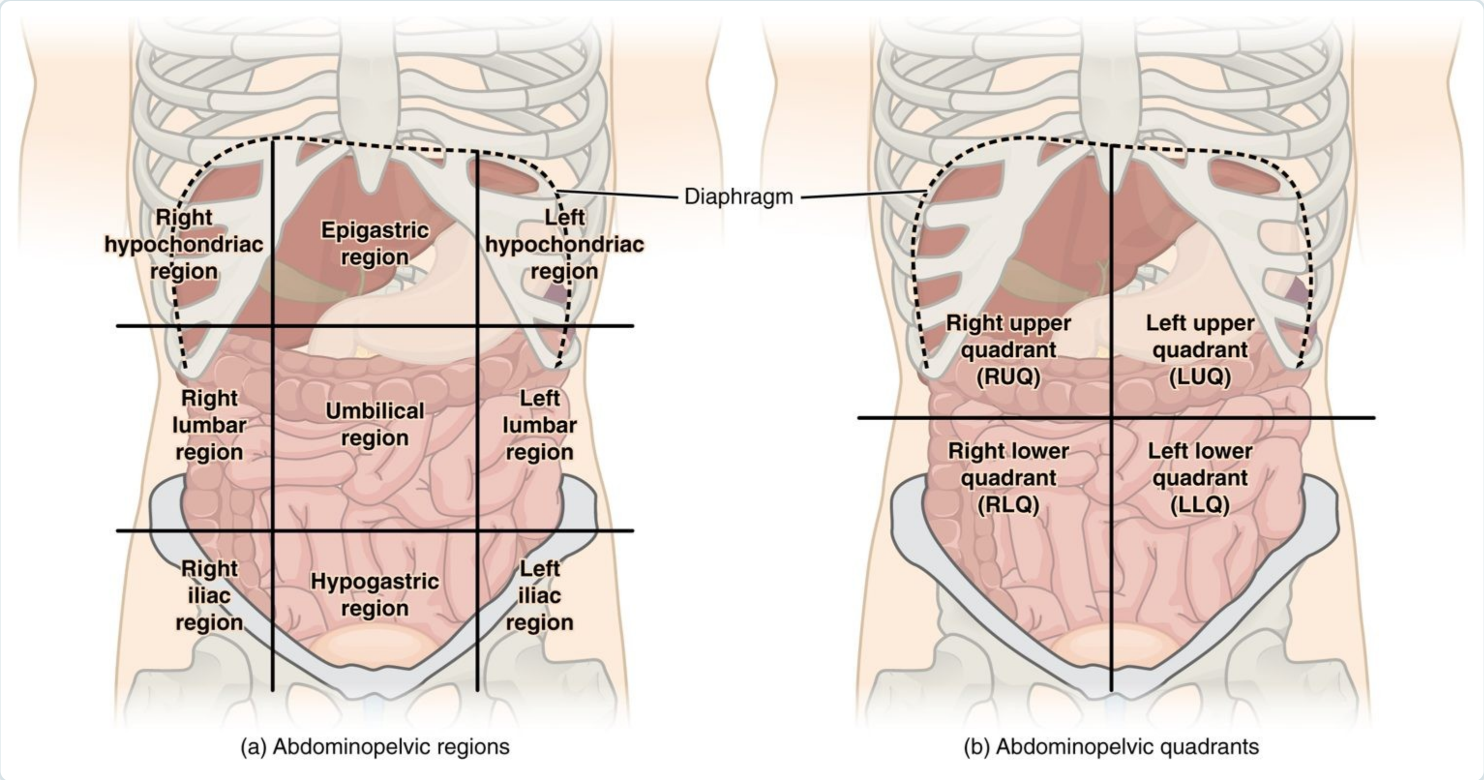
Body Cavities & Membranes



The dorsal and ventral body cavities

- Dorsal cavity: cranial and vertebral
- Ventral cavity: thoracic and abdominopelvic
- The diaphragm separates thorax and abdomen
- Cavities protect and organize the organs

Abdominal Regions & Quadrants



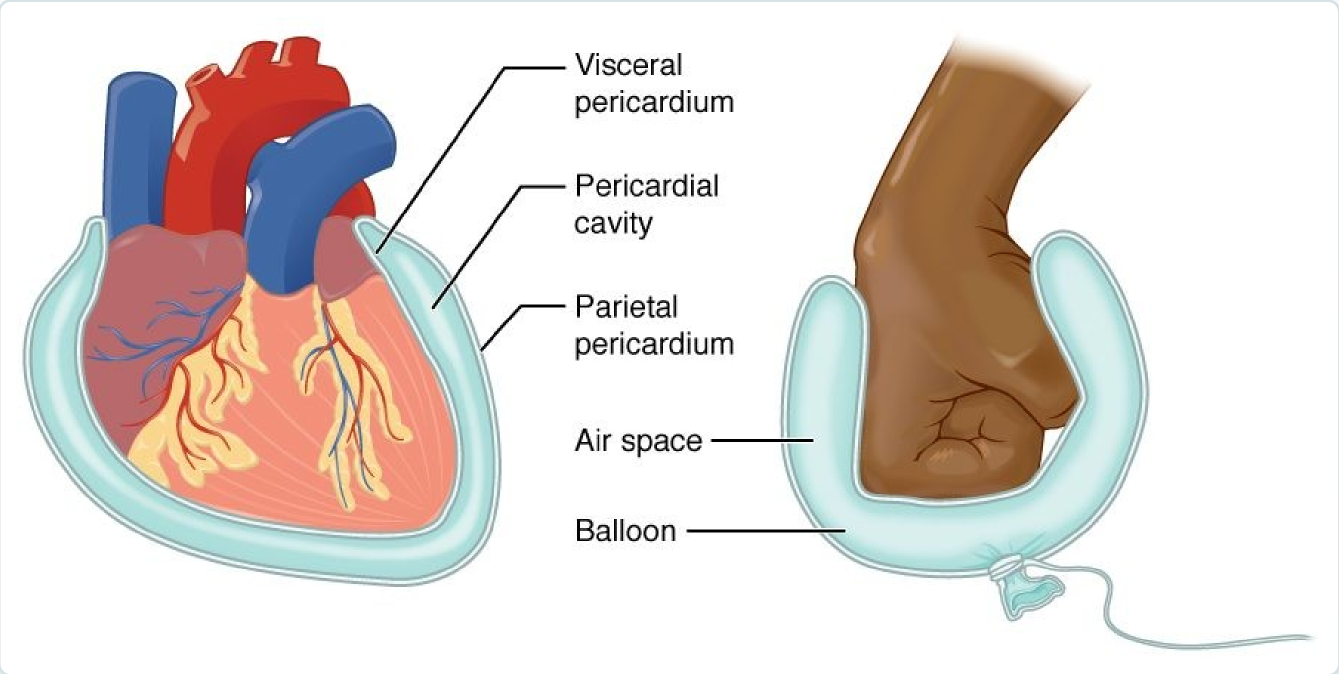
The nine abdominal regions and four quadrants

- Four quadrants: RUQ, LUQ, RLQ, LLQ
- Nine regions allow finer localization
- Used to describe pain and organ location
- RLQ pain → think appendicitis

SECTION 1.6

Serous Membranes

- Line cavities and cover the organs
- Parietal layer lines the wall; visceral covers the organ
- Serous fluid reduces friction
- Examples: pericardium, pleura, peritoneum



Serous membrane structure (balloon-and-fist analogy)

1.7

SECTION

Medical Imaging

Seeing inside the living body — radiography, CT, MRI, PET, and ultrasound.

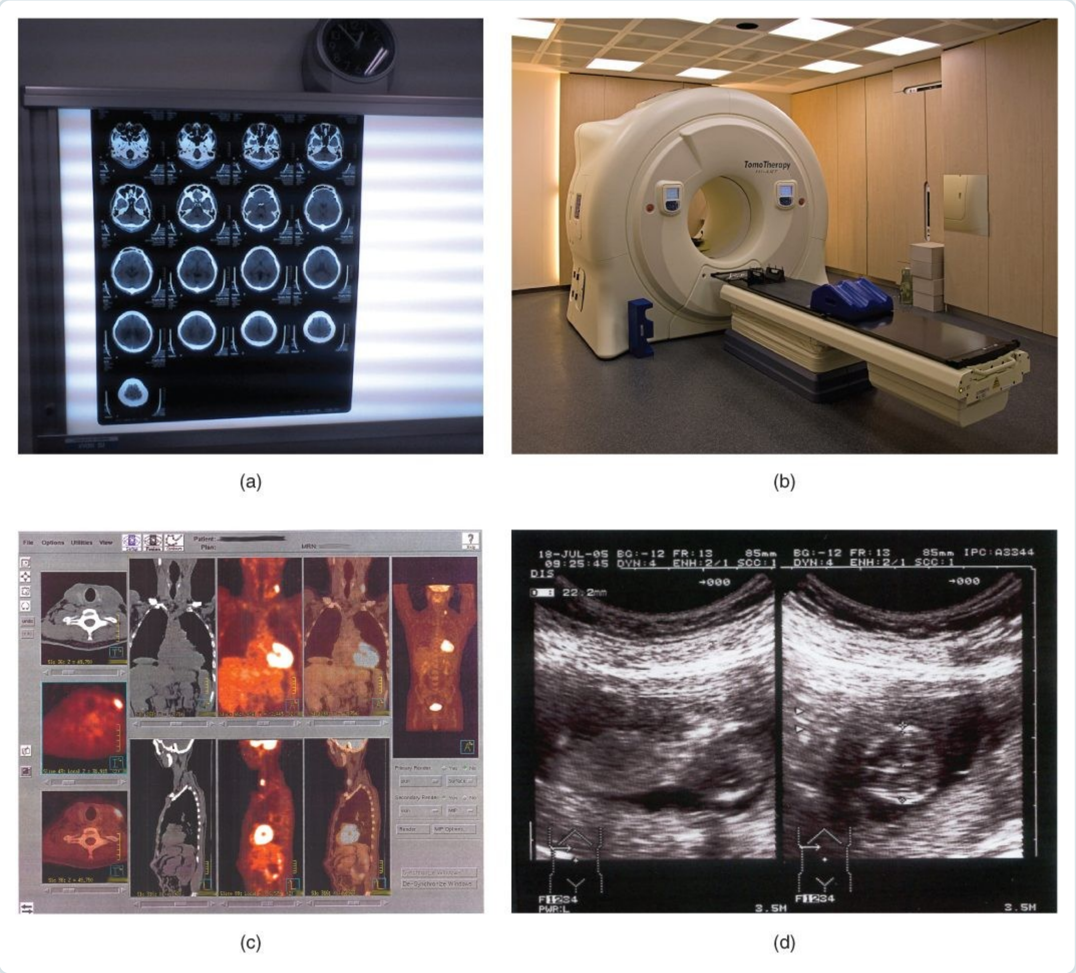
Medical Imaging — Radiography

A plain radiograph (X-ray) of the hand

- X-rays show dense structures like bone
- Quick, inexpensive, and widely available
- Limited soft-tissue detail
- First-line for suspected fractures

Imaging Modalities Compared

- CT: cross-sectional X-ray, excellent for trauma
- MRI: magnetic, superb soft-tissue detail
- PET: shows metabolic activity
- Ultrasound: safe, real-time, no radiation



Common imaging modalities and their uses

Key Takeaways

- Anatomy is structure and physiology is function — the two are inseparable.
- The body is organized in six levels, from the chemical level to the organism.
- Eleven organ systems work together to keep the body alive.
- Homeostasis, maintained mainly by negative feedback, keeps internal conditions stable.
- Anatomical terminology gives a precise, shared language for safe practice.

CLINICAL CONNECTIONS

Why the Basics Matter in Practical Nursing

Vital Signs

Temperature, BP, HR, and RR are homeostasis you measure every shift.

Assessment Language

Directional terms and quadrants make documentation precise and safe.

Abdominal Assessment

Quadrants and regions localize pain to the underlying organs.

Feedback Loops

Fever, blood-glucose, and BP control all follow feedback mechanisms.

Reading Imaging

Knowing the planes and modalities helps you interpret and explain studies.

Whole-Person Care

Systems are interdependent — one failing system affects the rest.